



BEYONDX

TECHNICAL PAPER 006

VSAM & Sequential Data Modernization

From Mainframe File Structures and COBOL Copybooks
to Relational and Cloud Data Architecture

VSAM • KSDS • ESDS • RRDS • Sequential Files • COBOL Copybooks
EBCDIC • COMP-3 • REDEFINES • Relational Data • Cloud Modernization

BEYONDX ENGINEERING POSITION

A VSAM or sequential file is not merely a file to be converted. The business data model must first be reconstructed from record layouts, copybooks, application logic, access patterns, encoding, and operational dependencies.

Version 4.0 | Final Publication Edition | September 2026

Stanley Reddy
Founder & Principal Modernization Architect
BEYONDX LLC

Executive Summary

VSAM and sequential datasets remain embedded in some of the most durable enterprise applications ever built. They support online transactions, batch processing, interfaces, extracts, temporary work files, master data, and decades of accumulated business logic. Their modernization challenge is deceptively simple: the stored bytes often do not carry enough metadata to explain themselves.

Unlike a relational database, where schemas expose columns, data types, keys, constraints, and indexes, the meaning of a VSAM or sequential record is frequently externalized into COBOL copybooks, PL/I structures, JCL, application programs, file definitions, operational procedures, and institutional knowledge. A successful modernization therefore begins with metadata reconstruction, not file conversion.

This paper examines KSDS, ESDS, RRDS, sequential files, COBOL copybooks, EBCDIC, packed decimal, REDEFINES, OCCURS, multiple record types, and the engineering decisions required to convert legacy record-oriented storage into explicit relational, cloud, object-storage, and analytical structures. It also addresses why enterprises modernize: operating economics, accessibility and integration, application coupling, workforce sustainability, analytics and AI enablement, and the ability to retire legacy dependencies with confidence.

CORE PRINCIPLE

The unit of VSAM modernization is not the physical record. It is the business entity encoded within the record.

A practitioner case study from BlueCross BlueShield of South Carolina illustrates the larger business outcome. In the early 2000s, a multi-year, application-by-application modernization journey progressed through membership, eligibility, real-time transaction processing, claims adjudication, and rapid Explanation of Benefits (EOB) generation. The lesson is important: real-time business capability emerged from progressively removing architectural dependencies, not from replacing a single file.

1. Why VSAM and Sequential Files Still Matter

Mainframe file architectures continue to matter because they are deeply coupled to business applications. VSAM datasets may be accessed by CICS online transactions, COBOL programs, utilities, and batch jobs. Sequential datasets are heavily used by JCL-driven batch processing for extracts, interfaces, sort/merge pipelines, reports, checkpoints, work files, and permanent business data.

The modernization driver is rarely that VSAM or sequential storage has suddenly stopped working. The pressure usually comes from changing expectations: continuous availability, real-time processing, API access, cloud analytics, easier integration, reduced batch windows, transparent metadata, and the ability to retire proprietary dependencies.

ENGINEERING INSIGHT

Modernization should not begin by asking, “How do we convert this file?” It should begin by asking, “Which business capabilities depend on this file, and where is the meaning of its data defined?”

Why Modernize VSAM and Sequential Data?

Analytics and AI readiness. Once appropriately governed legacy data is represented through accessible relational, object-storage, lakehouse, streaming, or analytical architectures, it can participate more naturally in modern BI, analytics, machine-learning, and AI ecosystems.

Workforce sustainability. Organizations must retain sufficient expertise in COBOL, VSAM, JCL, CICS, and associated technologies for as long as those systems remain operational. Modernization can reduce dependence on specialized implementations while preserving the business knowledge accumulated within them.

Application and operational coupling. Long-lived systems may tightly couple datasets with COBOL programs, CICS transactions, batch jobs, JCL, utilities, scheduling dependencies, and operational procedures. Modernization provides an opportunity to make those dependencies explicit and progressively reduce unnecessary coupling.

Accessibility and integration. VSAM does not inherently provide the SQL-based relational access model expected by many contemporary applications, APIs, analytics platforms, and data-engineering tools. Modernization can expose legacy business data through relational schemas, services, event streams, or other appropriate interfaces.

External and copybook-dependent metadata. VSAM and sequential datasets are not self-describing relational structures. Record definitions frequently reside externally in COBOL copybooks, PL/I structures, application programs, JCL, and related artifacts. Preserving the dataset without preserving and validating those definitions does not preserve authoritative business meaning.

Operating economics. Mainframe workloads involve specialized compute, storage, software, operations, and skills. For suitable workloads, distributed or cloud platforms may offer different consumption and scaling models. Cost reduction should be demonstrated through workload-specific analysis rather than assumed.

BEYONDx ENGINEERING POSITION

Modernization should not begin with the assumption that the mainframe is obsolete. It should begin by identifying which business capabilities, data assets, operational dependencies, economics, and strategic objectives justify change - and then selecting the appropriate target architecture.

2. Understanding VSAM Architecture

VSAM is a record-oriented access method on z/OS. CICS supports key-sequenced, entry-sequenced, and relative-record datasets. Each organization provides a different access model, and that access model influences how the data should be discovered and mapped to a target platform.

VSAM Type	Primary Organization	Typical Access	Modernization Consideration
KSDS	Records ordered by unique key	Key, browse, alternate index, relative location	Identify business key, alternate indexes, uniqueness, access paths
ESDS	Records retained in entry	Sequential or relative	Determine whether

	sequence	byte address	arrival order/RBA has business significance
RRDS	Records addressed by relative record number	Relative record number	Determine whether record position is a surrogate identifier or business rule

IBM documents that a KSDS identifies each record by a key at a predefined position and maintains an index relating keys to record locations. ESDS and RRDS differ in organization and addressing. These distinctions matter during migration because a physical access mechanism should not automatically become the logical data model of the target.

3. Sequential Files and Batch Processing

Sequential files are among the simplest mainframe storage structures physically, but they can participate in complex processing chains. A dataset may be created by one job step, sorted by another, enriched by a third, consumed by downstream applications, and retained for audit or reconciliation. Other files may exist only for minutes as temporary intermediate storage.

Modernization must classify each dataset before selecting a target. A permanent master or transactional file may belong in a relational database. A large extract or interchange file may fit cloud object storage. A temporary sort/work file may disappear entirely when the processing design changes.

File Role	Examples	Potential Target
System of record	Master/member/account records	Relational or cloud transactional database
Interface/extract	Partner feeds, outbound records	Object storage, managed file transfer, streaming
Batch intermediate	SORT/MERGE/work datasets	Pipeline-managed transient storage
Archive/audit	Historical snapshots, immutable extracts	Object/archive storage with retention controls

4. COBOL Copybooks: The External Data Contract

VSAM and sequential datasets generally do not expose relational-style metadata describing every field. Their metadata is commonly external to the physical data and dependent on the availability of authoritative COBOL copybooks, PL/I structures, application code, JCL, and related definitions. COBOL copybooks often provide the structural contract needed to interpret the bytes: field offsets, lengths, numeric representation, scale, repeating groups, alternative layouts, and record variants. If the correct copybook is missing, obsolete, or does not match the physical records, reconstruction becomes a discovery and reverse-engineering exercise.

BEYONDx ENGINEERING INSIGHT

The bytes may survive long after their metadata does. Preserving a VSAM or sequential dataset is not sufficient to preserve its business meaning. The corresponding copybooks, record definitions, encoding rules, and application context must also be discovered, validated, and preserved.

A simplified record might be defined as:

ILLUSTRATIVE COPYBOOK

```
01 CUSTOMER-RECORD.  
  05 CUSTOMER-ID    PIC 9(10).  
  05 CUSTOMER-NAME  PIC X(40).  
  05 CUSTOMER-TYPE  PIC X.  
  05 ACCOUNT-BALANCE PIC S9(11)V99 COMP-3.  
  05 CREATE-DATE    PIC 9(8).
```

A target relational design may interpret these fields as DECIMAL, VARCHAR, CHAR, DECIMAL with scale, and DATE. That mapping is not safe until actual data and program behavior are examined. For example, a PIC 9(8) date could represent YYYYMMDD, MMDDYYYY, a sentinel value, or a business-specific encoding.

5. Interpreting Legacy Bytes

Character conversion is only one part of the problem. Mainframe records may combine EBCDIC text, display numerics, packed decimal (COMP-3), binary computational fields, signed values, implied decimals, dates encoded as character or numeric fields, and application-specific flags.

Legacy Representation	Risk During Conversion	Target Treatment
EBCDIC text	Incorrect code-page conversion; embedded non-text bytes	Convert with known CCSID/code page and validate
COMP-3 / packed decimal	Sign nibble, precision, scale, invalid legacy values	Decode using copybook precision/scale; validate exceptions
COMP / binary	Byte order, signedness, compiler assumptions	Decode according to COBOL definition and platform rules
PIC 9 / X dates	Ambiguous format and sentinel values	Profile values; map only after semantics are confirmed
Flags/codes	Meaning exists only in programs or documentation	Create reference metadata and preserve business semantics

VALIDATION PRINCIPLE

Never validate only the number of records. Validate decoded values, numeric precision, signs, code-page conversion, key uniqueness, record-type distributions, financial totals, and business outcomes.

6. REDEFINES, OCCURS, and Multiple Record Types

COBOL record layouts can represent several logical structures within the same physical bytes. REDEFINES allows one storage area to be interpreted through different layouts. OCCURS represents repeating structures and may be fixed or dependent on another field. A single physical file may also contain multiple record types distinguished by a record-type code.

These constructs are normal in mainframe applications but require deliberate relational modeling. A REDEFINES structure should not automatically become one wide table containing every possible field. The engineering team must identify the discriminator, determine which logical entities exist, understand their cardinality, and decide whether the target should use separate tables, parent-child relationships, JSON/document structures, or another model.

CORE PRINCIPLE

Physical record equivalence does not imply business-entity equivalence. Flatten only when the target business model supports it.

7. Online and Batch Dependencies

A VSAM dataset may participate in both online and batch processing. CICS applications can read, add, update, rewrite, and delete VSAM records, while batch jobs and utilities may process the same or related datasets. The engineering challenge is therefore not that VSAM inherently lacks update capability; it is coordinating ownership, locking, sharing, recovery, batch processing, backup, and operational availability across the complete architecture.

IBM continues to describe products and operating patterns designed to reduce nightly batch windows and avoid taking CICS regions down for batch processing. This historical tension between online availability and batch processing is one reason many enterprises progressively moved transactional workloads toward database-centered architectures.

IMS/DC or other transaction managers can add another dependency layer. Discovery should trace not only which program reads a file, but which transaction, screen, message, batch job, scheduler, interface, and downstream process depends upon the resulting data.

8. Practitioner Case Study: BlueCross BlueShield of South Carolina

In the early 2000s, BlueCross BlueShield of South Carolina pursued a major technology modernization journey that moved critical healthcare capabilities from predominantly batch-oriented processing toward increasingly real-time services.

The company's official historical archive documents that BlueCross implemented real-time claims adjudication for commercial business in 2001 and describes the company as the first in the nation to offer doctors the ability to file claims via the internet. The archive also notes that one year later BlueCross became the first Blue Plan to fully implement electronic processing of HCFA 1500 and dental claims.

From the author’s firsthand project experience, this was not a single-system conversion. It required approximately three to four years of progressive, application-by-application modernization. Capabilities were advanced in sequence, beginning with membership and insurance eligibility validation and ultimately extending to real-time claims processing and adjudication, followed by real-time generation of the Explanation of Benefits (EOB).

Modernization Progression
Membership → Insurance Eligibility Validation → Real-Time Claims Processing → Claims Adjudication → Real-Time EOB Generation

Providers and institutions could continue to submit claims through batch channels while increasingly using online capabilities. As providers became accustomed to entering procedure and related information closer to the point of service, more processing occurred during the business transaction rather than waiting for the nightly cycle.

According to the author’s project recollection, an early target was to adjudicate a claim in approximately four minutes. The first deployment was able to adjudicate claims in less than one minute. Following adjudication, the Explanation of Benefits (EOB) for affected workflows could be generated in seconds, based on the author's project recollection. These performance observations are practitioner recollections and are presented separately from the independently documented corporate timeline.

ENGINEERING LESSON

Real-time adjudication was not created by replacing one VSAM file. It emerged from progressively modernizing applications, data storage, data access, transaction processing, interfaces, operational procedures, and dependencies while preserving ongoing business operations.

9. Leadership Insight: Building the Wall One Section at a Time

An ancient leadership analogy from the biblical Book of Nehemiah, Chapter 3, provides a useful way to think about large modernization programs. The rebuilding of Jerusalem's wall was divided into manageable sections and distributed across priests, craftsmen, merchants, rulers, families, and people from surrounding communities. Several groups repaired sections near their own homes or areas of responsibility. The result was distributed ownership of one shared mission rather than dependence on a single labor force.

The analogy is organizational, not technical: a complex objective becomes achievable when work is decomposed into feasible units, responsibility is placed close to knowledge and ownership, and every completed section is continuously connected to the larger outcome.

Nehemiah 3 principle	Modernization application
Feasible sections	Decompose an enterprise transformation into bounded capabilities that teams can execute and

	validate.
Responsibility close to the work	Place ownership with the business and engineering teams that understand each capability and its dependencies.
Broad participation	Bring application, data, infrastructure, operations, security, testing, architecture, and business disciplines into the mission.
Shared ownership	Each team owns a section, but every section must connect to the same target operating model.
Continue despite resistance	Resolve or route around organizational blockers without allowing one non-participating group to immobilize the program.
Integrate the sections	A modernized component has limited value if adjacent applications, interfaces, data, and operations cannot work with it.

Nehemiah 3:12 also records Shallum repairing his section with his daughters, while Nehemiah 3:5 records that some nobles of Tekoa did not participate. The narrative therefore captures both broad participation and the practical reality that not every stakeholder will contribute equally. Nehemiah 6:15 records completion of the wall in 52 days - an illustration of coordinated parallel execution, not a suggested timetable for technology modernization.

The parallel to the BlueCross modernization experience is the pattern of execution. Membership, eligibility, online transactions, claims adjudication, and EOB generation were not isolated destinations. Different teams progressively modernized their respective sections until those capabilities connected and supported the larger real-time operating model.

BEYONDX ENGINEERING PRINCIPLE

Modernize where the knowledge lives. Divide the mission into feasible sections, give ownership to the people closest to each capability, continuously connect and validate the sections, and keep every team aligned to the same enterprise outcome.

10. Reconstructing the Business Data Model

The target schema should be designed from discovered business entities and relationships, not merely from byte offsets. A disciplined discovery process correlates copybooks with actual records, programs, JCL, file definitions, keys, alternate indexes, transaction flows, and business documentation.

Discovery Step	Key Questions
Inventory	Which VSAM/sequential datasets exist? Who creates, reads, updates, and deletes them?
Layout reconstruction	Which copybook/version applies? Are there multiple layouts, REDEFINES, OCCURS, variable records?
Data profiling	What values actually exist? Are there invalid packed

	decimals, unexpected encodings, sentinels, duplicates?
Dependency analysis	Which CICS/IMS transactions, COBOL programs, JCL jobs, utilities, interfaces, and reports depend on the data?
Business modeling	Which physical fields represent entities, keys, relationships, codes, history, and events?
Target design	Which data belongs in relational tables, object storage, streaming, or transient pipelines?

11. Conversion to Relational and Cloud Structures

Once the source semantics are understood, the target can make metadata explicit. Keys can become primary or unique keys, repeating groups can become child tables where appropriate, code fields can gain reference metadata, and relationships previously enforced through program logic can be represented through constraints or services.

Legacy Pattern	Possible Modern Target
KSDS keyed master	Relational table with PK/unique key and supporting indexes
Alternate index	Secondary index or alternate access path
ESDS event/history stream	Append-oriented relational/event table or object/streaming architecture
RRDS positional lookup	Table with explicit logical identifier; preserve RRN only if semantically required
REDEFINES record types	Subtype tables, parent-child design, or document structure based on semantics
OCCURS repeating group	Child table, array/document field, or normalized relationship
Sequential interchange file	Object storage, managed transfer, ingestion pipeline, or stream

Modernization does not require every legacy file to become a database table. The correct target follows the business role of the data.

12. Migration, Coexistence, and Validation

For high-value operational systems, migration should support controlled coexistence. Online and batch workloads may need to continue while datasets are extracted, decoded, transformed, loaded, reconciled, and cut over. The migration architecture should explicitly address how changes are captured during the transition and how rollback or fallback will operate.

- Establish authoritative copybook and record-layout versions before conversion.
- Profile source bytes before applying target data-type assumptions.
- Convert encoding and numeric representations deterministically.
- Validate record counts by record type, key ranges, dates, and business partitions.
- Reconcile financial/control totals and representative business transactions.
- Test both online and batch pathways during coexistence.
- Measure target behavior under realistic concurrency and volume.
- Preserve evidence sufficient to support final legacy retirement.

BEYONDX PRINCIPLE

Discovery is not a phase—it is a discipline. Validation is not a milestone—it is continuous.

13. AI-Assisted Mainframe Modernization

VSAM and sequential modernization is particularly well suited to AI-assisted engineering because business meaning is distributed across external metadata, COBOL copybooks, application programs, JCL, CICS definitions, utilities, documentation, and operational knowledge. Generative AI can accelerate discovery, migration, validation, and documentation when it is used inside an engineering process with authoritative-source validation, security controls, and human review.

ChatGPT and OpenAI capabilities can serve as modernization engineering copilots - assisting with legacy-code comprehension, copybook interpretation, dependency analysis, target-schema design, conversion logic, test generation, reconciliation, and technical documentation. Where organizations use AI with proprietary or regulated source material, the selected deployment must comply with enterprise security, privacy, data-handling, and authorization requirements.

- Copybook analysis - interpret PIC clauses, COMP/COMP-3, OCCURS, REDEFINES, signs, implied decimals, nested structures, and candidate record lengths.
- Metadata reconstruction - correlate datasets with candidate copybooks, JCL DD statements, file-control definitions, application code, field usage, and representative records.
- Dependency discovery - trace COBOL programs, CICS transactions, IMS/DC components, JCL jobs, utilities, interfaces, reports, and downstream consumers that read or modify a dataset.
- Target-model assistance - propose relational tables, keys, relationships, constraints, indexes, cloud storage placement, APIs, and conversion mappings for engineering review.
- Conversion and test assistance - generate candidate parsing logic, EBCDIC and packed-decimal handling, SQL, ETL code, reconciliation queries, boundary tests, and validation scenarios.
- Documentation - transform discovered layouts, lineage, dependencies, decisions, mapping rules, and validation evidence into maintainable engineering documentation.

ChatGPT as a Modernization Engineering Copilot

ChatGPT can help experienced engineers investigate unfamiliar COBOL and JCL, compare copybook variants, explain legacy access logic, develop candidate target schemas and conversion routines, generate validation SQL and test cases, and turn discovery findings into usable engineering documentation. Its value is acceleration and correlation - not autonomous determination of business truth.

AI-generated interpretation is not authoritative source metadata.

For enterprise modernization, AI use should follow approved security and data-governance controls. OpenAI states that business data from ChatGPT Business, Enterprise, Edu, and the API platform is not used to train models by default. Organizations should still apply their own authorization, retention, access-control, classification, and regulatory requirements before submitting proprietary, sensitive, or regulated data to any AI system.

When an authoritative copybook is missing, AI may help infer candidate structures by correlating available bytes, program logic, JCL, field usage, record samples, and historical documentation. That inference is a hypothesis, not authoritative metadata. It must be validated against actual records, application behavior, business rules, control totals, and subject-matter expertise before it becomes part of a production migration specification.

BEYONDX ENGINEERING PRINCIPLE

AI accelerates modernization - but it does not replace engineering judgment. Generated interpretations must be validated against source data, application behavior, business rules, security requirements, and subject-matter expertise before they become migration specifications.

BEYONDX ENGINEERING PRINCIPLE

AI accelerates discovery. AI accelerates conversion. AI accelerates validation. Engineering judgment establishes correctness.

14. Decommissioning: The Real Definition of Success

A file migration is incomplete while production applications, batch jobs, interfaces, recovery procedures, or operational teams still depend upon the legacy datasets. Decommissioning requires evidence that all authoritative reads and writes have moved, reconciliations are complete, retention obligations are satisfied, fallback requirements have expired, and the legacy storage can be retired without business impact.

Data migration creates a copy. Dependency elimination enables decommissioning.

The goal is not to preserve VSAM semantics forever on a newer platform. The goal is to preserve business capability while replacing externally defined and application-dependent metadata, encoded record structures, and tightly coupled access patterns with explicit, supportable architecture.

BEYONDX ENGINEERING POSITION

Modernization is successful only when the business operates with confidence on the target platform and the legacy dependency can be retired.

15. BeyondX VSAM & Sequential Modernization Checklist

Phase	Essential Evidence
Discover	Dataset inventory; copybooks; JCL; CICS/IMS dependencies; program usage; file definitions
Decode	CCSID/EBCDIC rules; COMP/COMP-3; signs; dates; REDEFINES; OCCURS; variable records
Model	Business entities; keys; relationships; history; record-type semantics; target placement
Migrate	Extraction; conversion; loading; coexistence/change strategy; fallback
Validate	Counts; totals; decoded values; keys; transactions; online/batch results; performance
Retire	Dependency closure; retention/archive; operational sign-off; legacy decommissioning
AI Assist	Copybook/code analysis; dependency correlation; candidate mappings; generated tests and reconciliation - all subject to engineering validation

References and Source Notes

Biblical leadership analogy: Nehemiah 3:5, 3:12, 3:23, 3:28, 3:30; Nehemiah 6:15. Scripture is used as a leadership analogy and not as technical evidence.

1. BlueCross BlueShield of South Carolina, “Technology Takes Off,” celebrate blueSC historical archive. Documents the 2001 real-time claims adjudication milestone and subsequent electronic claims-processing milestones.
<https://www.celebratebluesc.com/technology-takes>
2. IBM, CICS Transaction Server for z/OS: Developing CICS Applications. VSAM data sets: KSDS, ESDS, and RRDS; CICS file-control access and record operations. https://www.ibm.com/docs/en/SSJL4D_6.x/pdf/application-programming_pdf.pdf
3. IBM, z/OS DFSMS Using Data Sets. Descriptions of VSAM organizations and access characteristics.
https://www.ibm.com/docs/en/SSLTBW_3.1.0/pdf/idad400_v3r1.pdf
4. IBM, CICS Batch Application Control for z/OS. Describes reducing nightly batch windows and avoiding CICS shutdown for batch processing. <https://www.ibm.com/products/batch-application-processing>
5. IBM COBOL support documentation. Packed-decimal (COMP-3), zoned-decimal, and binary validation considerations.
<https://www.ibm.com/support/pages/apar/PI71625>

PRACTITIONER SOURCE NOTE

The BlueCross BlueShield of South Carolina case study combines an independently documented corporate timeline with Stanley Reddy’s firsthand recollection of the multi-year engineering effort. Project-duration, sequencing, initial performance target, sub-one-minute adjudication, and end-of-day performance observations are identified as practitioner experience rather than claims derived from the public archive.

6. OpenAI, Business Data Privacy, Security, and Compliance. OpenAI states that business data from ChatGPT Business, Enterprise, Edu, and the API platform is not used to train models by default. <https://openai.com/business-data/>
7. OpenAI, Running Codex Safely at OpenAI. Describes governance, technical boundaries, human approvals, and telemetry for coding-agent workflows. <https://openai.com/index/running-codex-safely/>

BEYONDX LLC

Technology with Purpose. Opportunity with Impact.

Modernize first. Transform second. Optimize continuously.